# Mock Exam — Statistics

2026-01-21

**Instructions**

- Answer directly in this file and knit to HTML or PDF at the end.
- Provide both **R output** and **short written interpretation** where requested.
- Total: **100 points**.

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## Part A — R Basics and Probability Theory (40 points)

### A1. Data types and factors (7 points)

Create the following object and answer the questions.

```r
x <- c(1, 2, 2, 3, 3, 3)
g <- factor(c("A","B","A","B","B","A"))
df <- data.frame(x = x, g = g)
str(df)
```

```
## 'data.frame':    6 obs. of  2 variables:
##  $ x: num  1 2 2 3 3 3
##  $ g: Factor w/ 2 levels "A","B": 1 2 1 2 2 1
```

1. What does one **row** in `df` represent?

   each row represents a value of x mapped to the corresponding value of g at the same index, representing a value mapped to a group

2. How many observations are in group `A` and in group `B`?

   A:3 B:3

3. Compute the **mean of `x`** separately for each group.

   ```r
   aggregate(df[1], list(df$g), mean)
   ```

   ```
   ##   Group.1        x
   ## 1       A 2.000000
   ## 2       B 2.666667
   ```

## A2. Monte Carlo simulation (8 points)

Let $\Omega = \{1, 2, \ldots, 100\}$. Draw $N = 50{,}000$ values uniformly from $\Omega$ (with replacement). Estimate:

$$P(\text{multiple of 3} \mid \text{multiple of 3 or 5})$$

1. Compute the Monte Carlo estimate in R.

```r
omega <- 1:100

drawsize <- 50000

drawN <- sample(omega, drawsize, replace = TRUE)



count <- 0
maxcount <- 0
for(draw in drawN){

  if(draw %% 3 == 0 || draw %% 5 == 0){
    maxcount <- maxcount + 1

    if(draw %% 3 == 0){
        count <- count + 1
    }

  }

}
Pmul35 <- count / maxcount
Pmul35
```

```
## [1] 0.7034668
```

2. Compute the exact value by counting and compare.
   div3 <- floor(100 / 3) div5 <- 100 / 5 div15 <- floor(100 / 15)
   exactPmul35 <- div3 / (div3 + div5 - div15)

## A3. Sampling from a non-normal distribution (25 points)

The goal of this exercise is to illustrate the **Central Limit Theorem** using simulation.

Let $X \sim \text{Exponential}(\lambda = 1)$.

1. Draw $N = 10{,}000$ independent samples of size $n = 5$, $n = 30$, and $n = 100$.

```r
drawsize <- 10000

X <- rexp(drawsize, 1)


n5 <- sample(X, 5)
n30 <- sample(X, 30)
n100 <- sample(X, 100)
```

2. For each $n$, compute the **sample mean**.

```r
mean(n5)
```

```
## [1] 0.9099417
```

```r
mean(n30)
```

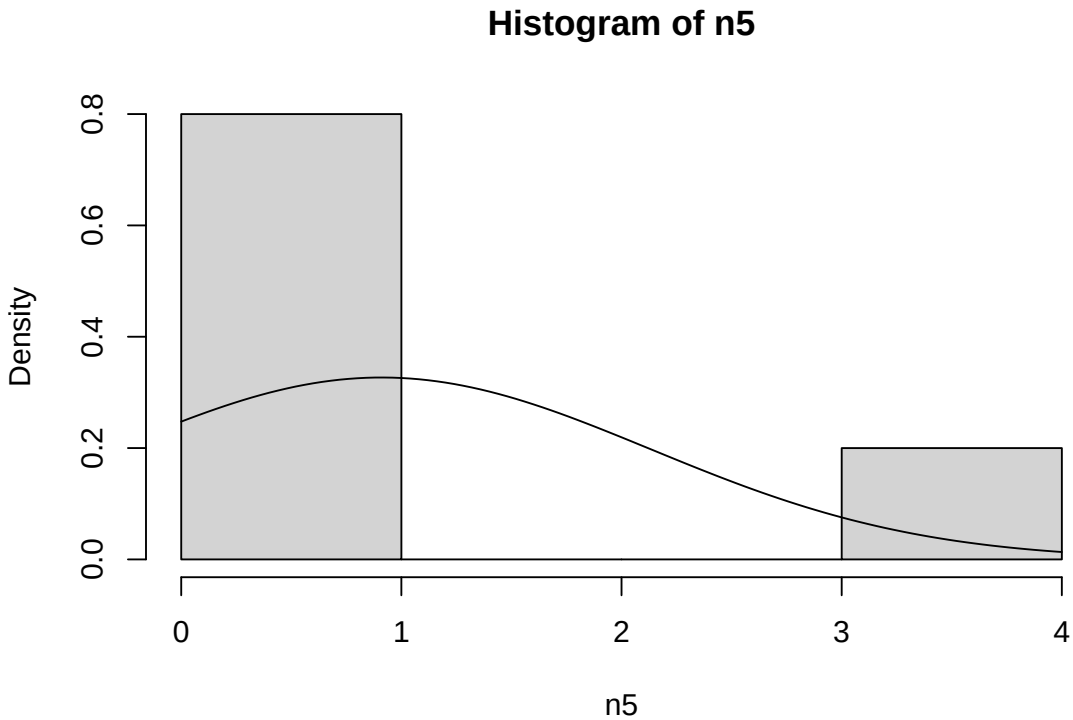
```
## [1] 0.9321358
```

```r
mean(n100)
```

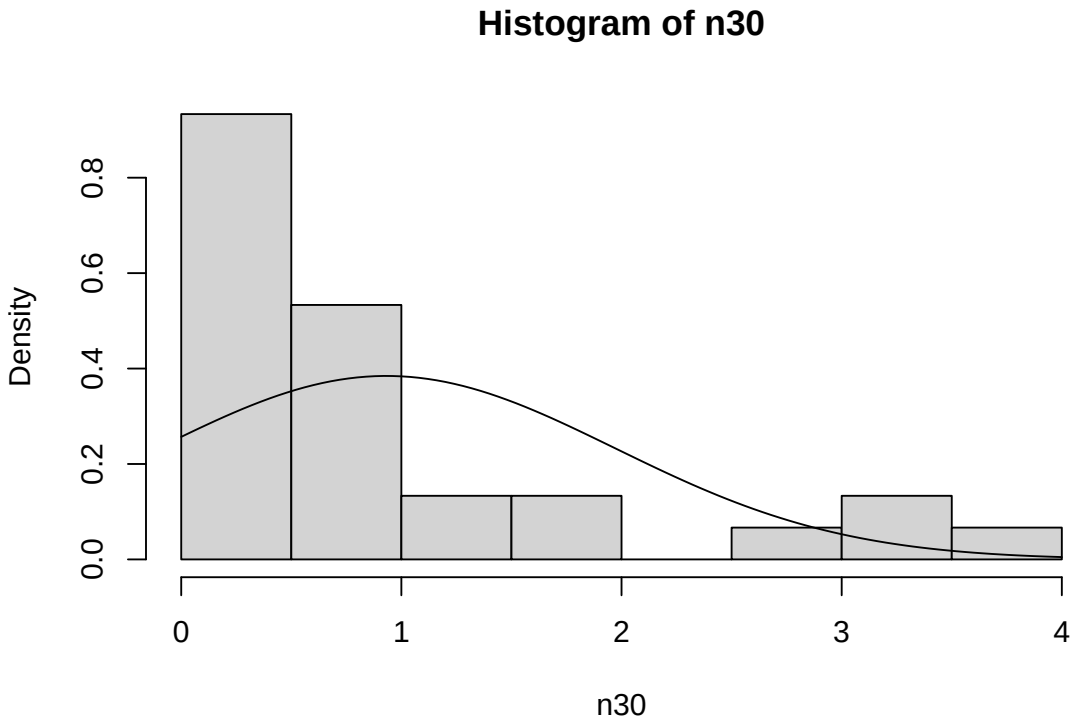
```
## [1] 1.114256
```

3. For each $n$, create a **histogram of the sample means** and overlay a **normal density** with the appropriate mean and variance.

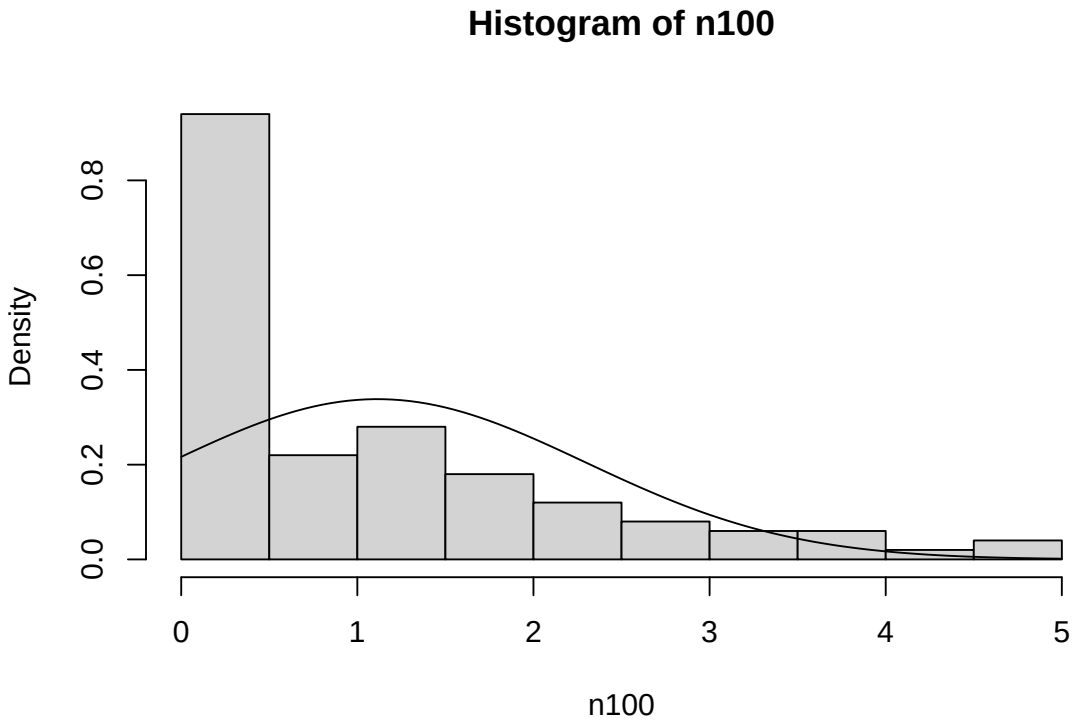
```r
hist(n5, probability = TRUE)
curve(
  dnorm(x, mean = mean(n5), sd = sd(n5)),
  add=TRUE
)
```




```r
hist(n30, probability = TRUE)
curve(
  dnorm(x, mean = mean(n30), sd = sd(n30)),
  add=TRUE
)
```




```
hist(n100, probability = TRUE)
curve(
  dnorm(x, mean = mean(n100), sd = sd(n100)),
  add=TRUE,
)
```




4. For each sample size $n$, define the standardized mean

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}},$$

where $\mu$ and $\sigma$ are the true mean and standard deviation of the exponential distribution.

a) Compute $Z_n$ for $n = 5, 30, 100$.

```
Z5 <- (mean(n5) - 1) / (1 / sqrt(5))
Z5
```

```
## [1] -0.2013764
```

```
Z30 <- (mean(n30) - 1) / (1 / sqrt(30))
Z30
```

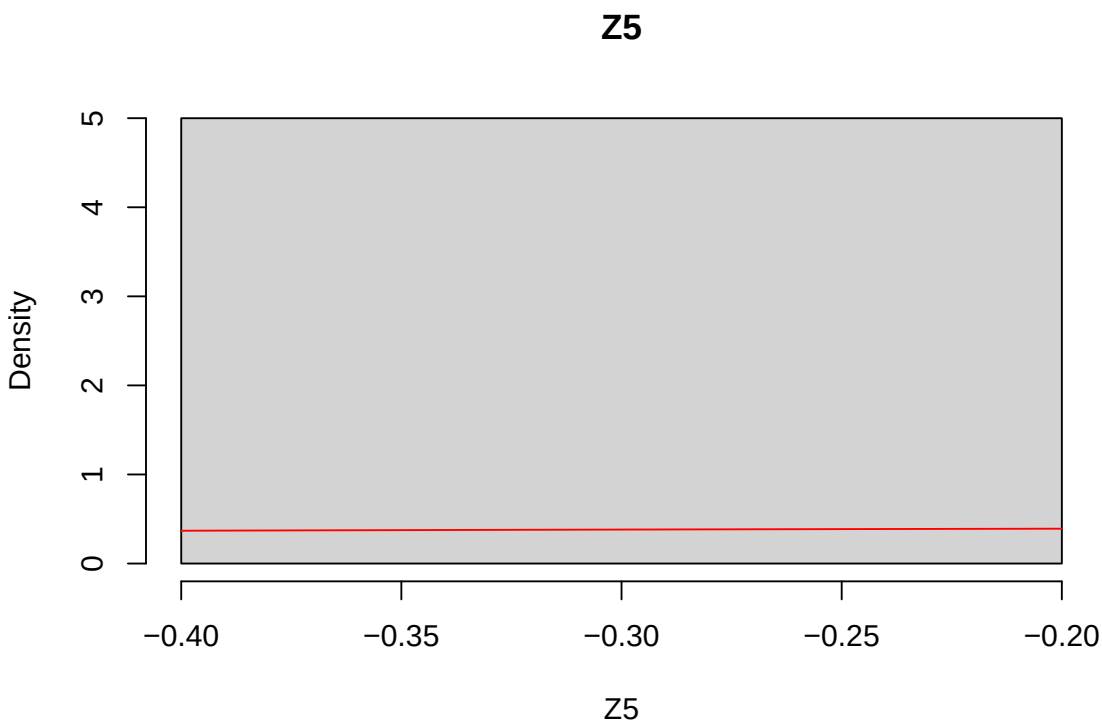
```
## [1] -0.3717078
```

```
Z100 <- (mean(n100) - 1) / (1 / sqrt(100))
Z100
```

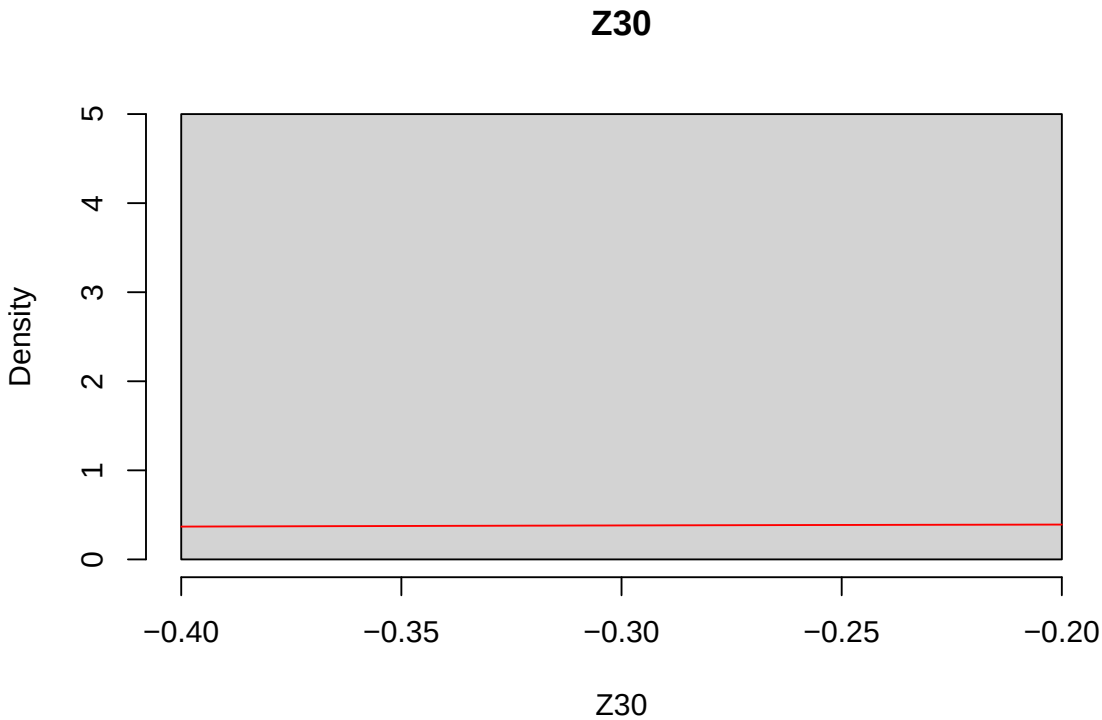
```
## [1] 1.142562
```

b) Create histograms of $Z_n$ and compare them to the standard normal distribution.

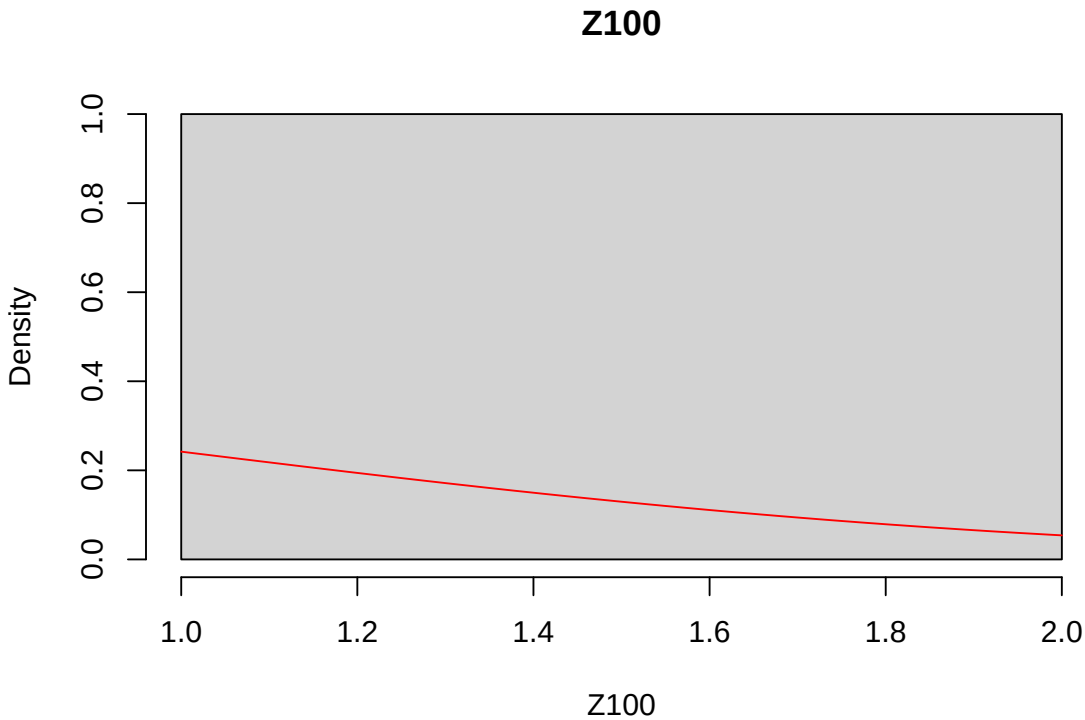
```r
hist(Z5, probability = TRUE, main = "Z5")
curve(dnorm(x), add = TRUE, col = "red")
```




```r
hist(Z30, probability = TRUE, main = "Z30")
curve(dnorm(x), add = TRUE, col = "red")
```




```r
hist(Z100, probability = TRUE, main = "Z100")
curve(dnorm(x), add = TRUE, col = "red")
```




Answer the following:

c) How does the distribution of $\bar{X}_n$ change as $n$ increases?

the dist. of X_n gets more dense the higher n is, e.g. it gets closer to lamdba (1)

d) Why does the CLT apply even though the original data are not normally distributed?

CLT still works because for high sample sizes (10000), every distribution approaches normal distribution

e) What is the purpose of standardization in the CLT?

standardization makes it possible to compare Z_n's of different n's and lambda's, as they normally have completely different means and distributions.

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## Part B — Descriptive statistics (45 points)

We use `birthwt` from the MASS package.

```r
library(MASS)
library(dplyr)
library(ggplot2)
library(forcats)

data(birthwt)

bwt <- birthwt %>%
  mutate(
    smoke = factor(smoke, levels = c(0,1), labels = c("No","Yes")),
    ht    = factor(ht, levels = c(0,1), labels = c("No","Yes")),
    ui    = factor(ui, levels = c(0,1), labels = c("No","Yes")),
    low   = factor(low, levels = c(0,1), labels = c("No","Yes"))
  )
```

## B1. Variable overview (6 points)

1. In your own words: what does one row of `bwt` represent?

   one row of bwt represents data from one birth

2. List **two categorical** and **two numeric** variables in `bwt`.

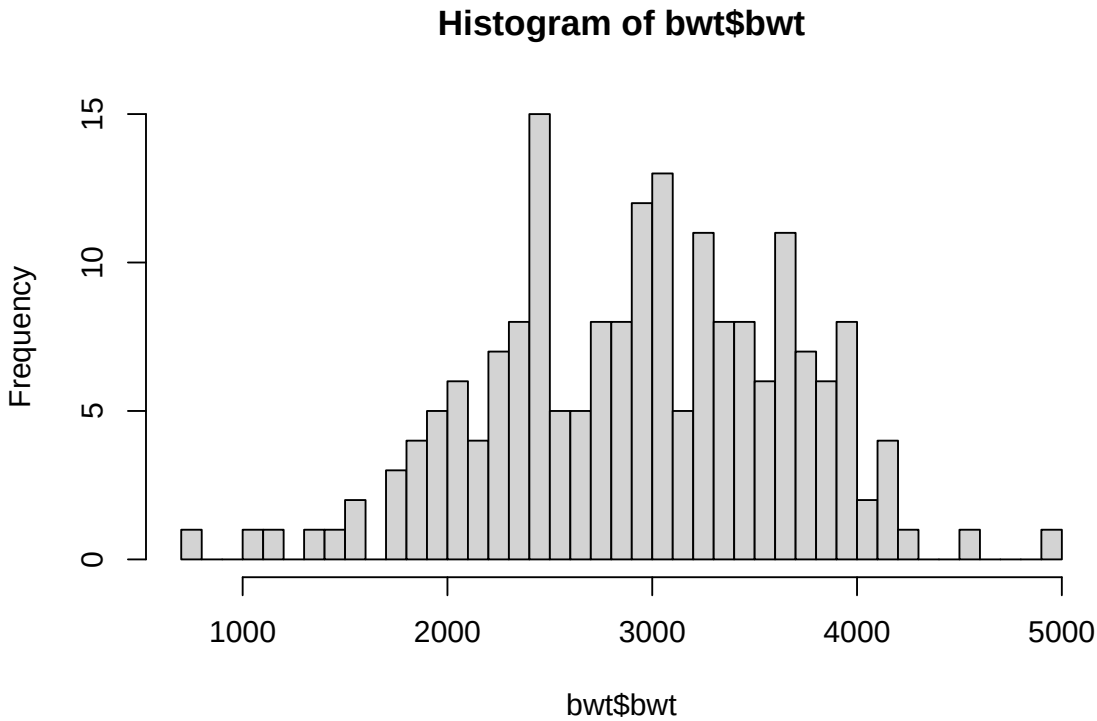
   cat.: low / smoke / race / ht / ui num.: age / lwt / ptl / ftv / bwt

## B2. Histograms + ECDF (10 points)

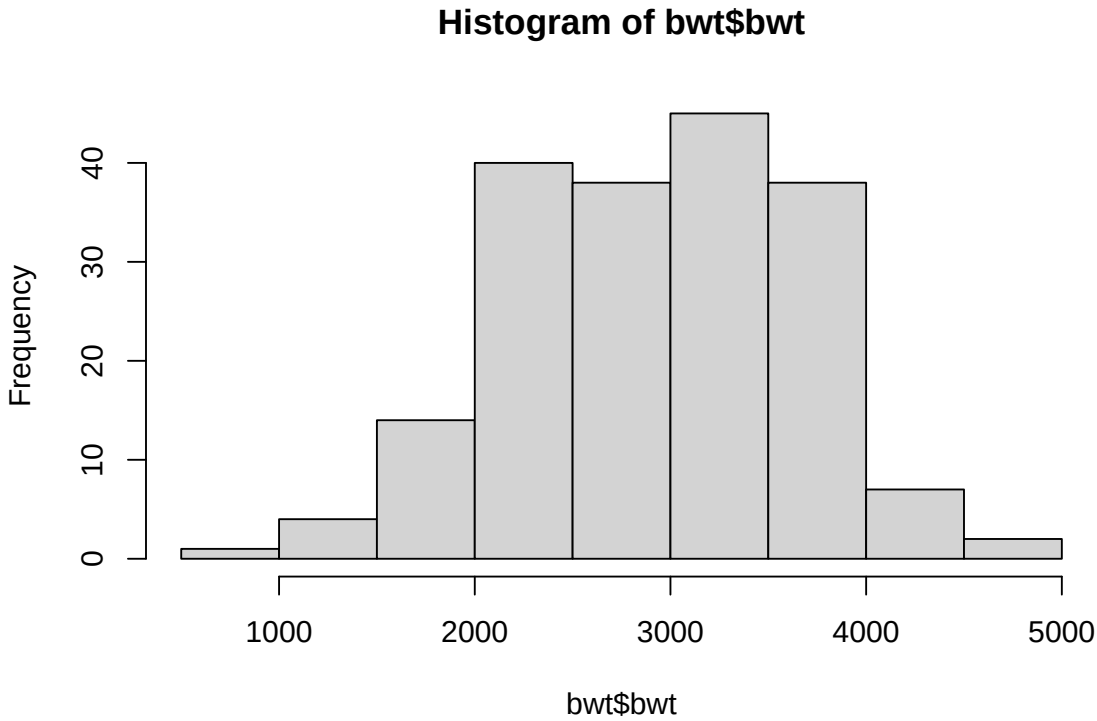
For the numeric variable `bwt$bwt` (birth weight in grams):

1. Create two histograms with different bin widths (e.g. 150 and 400).

   ```r
   hist(bwt$bwt ,breaks=5000 / 150)
   ```

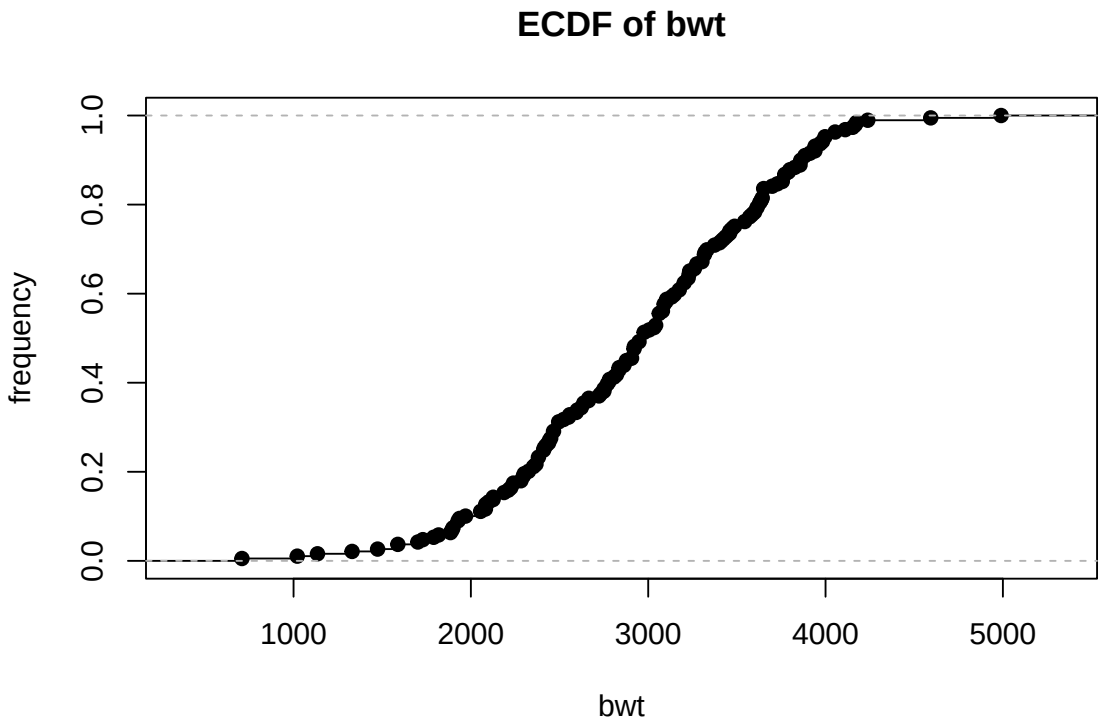



```r
hist(bwt$bwt ,breaks=5000 / 400)
```



2. Create an ECDF plot.

```
plot(ecdf(bwt$bwt), xlab = "bwt", ylab = "frequency", main = "ECDF of bwt")
```




3. From the ECDF (or in R), estimate the proportion of babies with birth weight $\leq$ 2500 g. Write a sentence or two interpreting 1) how bin width affects the histogram and 2) what you see in the ECDF.

```
'
My estimate: 0.3 -> 30%
1)Higher bin witdh makes the graph more smooth, but makes it harder to find outsiders with
2) the ECDF shows that most babies weigh 3kg or more after birth, and that the weight stag
for low values and ~5kg for high values. Most babies weigh between ~3kg and ~5kg.
'
```

```
## [1] "\nMy estimate: 0.3 -> 30%\n1)Higher bin witdh makes the graph more smooth, but mak
```

*Write your answer here.*

## B3. Group comparison: smokers vs non-smokers (9 points)

1. Create a **boxplot** of `bwt` by `smoke`.

```r
boxplot(bwt$bwt ~ bwt$smoke,
    xlab = "Smoking status", ylab = "Birth weight",
    names = c("Non Smoker", "Smoker"),
    main= "Zaza plot")
```

2. Report the median birth weight for each group.

```r
#Median weight for non-smokers
median(bwt$bwt[bwt$smoke == "No"])
```

```
## [1] 3100
```

```r
#Median weight for smokers
median(bwt$bwt[bwt$smoke == "Yes"])
```

```
## [1] 2775.5
```

3. In 3–4 sentences, summarize how smoking relates to birth weight using (i) the boxplot and (ii) the medians.

```
'
Like seen in the boxplot and the median values, smoking during pregnancy can
substantially reduce the birth weight of the child. Babies from smoking mothers
```

```
also have a smaller interval for their birth weight, following the median of ~2.8kg
much more closeley then the non-smoker-babies follow the median of 3.1kg.
'
```

```
## [1] "\nLike seen in the boxplot and the median values, smoking during pregnancy can\nsu
```

*Write your answer here.*

### B4. Contingency table + chi-square (10 points)

Consider the relationship between `smoke` and `low`.

1. Create the contingency table.

```r
table(bwt$smoke , bwt$low)
```

```
##
##       No Yes
##   No  86  29
##   Yes 44  30
```

2. Compute **row-wise proportions** (within smoking status).

```r
prop.table(table(bwt$smoke , bwt$low))
```

```
##
##              No       Yes
##   No  0.4550265 0.1534392
##   Yes 0.2328042 0.1587302
```

3. Perform a chi-square test of independence and state the null hypothesis in words.

```r
#H0: if p > 0.05, bwt$smoke and bwt$low are not related
chisq.test(bwt$smoke, bwt$low)
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  bwt$smoke and bwt$low
## X-squared = 4.2359, df = 1, p-value = 0.03958
```

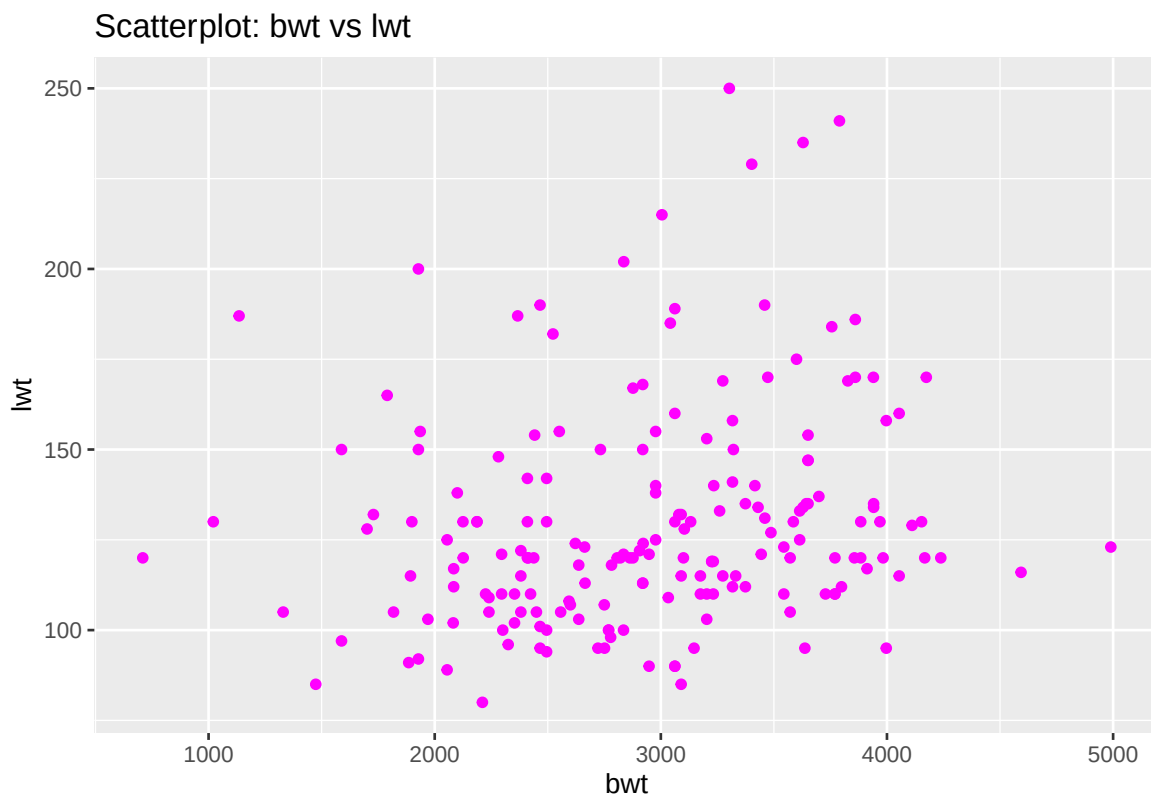
```r
# p) 0.03958 -> H0 gets thrown away
```

### B5. Scatterplot + correlation (10 points)

Study the relationship between maternal weight `lwt` and birth weight `bwt`.

1. Make a scatterplot (`bwt` vs `lwt`).

```
ggplot(bwt, aes(x = bwt, y = lwt)) +
  geom_point(colour = "magenta") +
  labs(title = "Scatterplot: bwt vs lwt")
```




2. Compute the correlation and interpret the sign and magnitude.

```
cor(bwt$bwt, bwt$lwt)
```

```
## [1] 0.1857333
```

```
# -> 0.1857333
# sign -> positive -> the higher the $lwt, the higher the $bwt
# magnitude ->  ~0.18 -> weak correlation
```

---

# Part C — Inferential statistics (15 points)

## C1. Fit and interpret a linear model (15 points)

Fit the model

$$\texttt{bwt} \sim \texttt{lwt}.$$

1. Fit the model and print the summary.

```r
bwtlm <- lm(bwt$bwt ~ bwt$lwt)

summary(bwtlm)
```

```
##
## Call:
## lm(formula = bwt$bwt ~ bwt$lwt)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2192.12  -497.97    -3.84   508.32  2075.60
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 2369.624    228.493  10.371   <2e-16 ***
## bwt$lwt        4.429      1.713   2.585   0.0105 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 718.4 on 187 degrees of freedom
## Multiple R-squared:  0.0345, Adjusted R-squared:  0.02933
## F-statistic: 6.681 on 1 and 187 DF,  p-value: 0.0105
```

2. Write down the estimated regression equation.

### f: estimated regression equation; f(x) = 4.429x + 2369.624

3. Interpret the slope in context (one sentence).

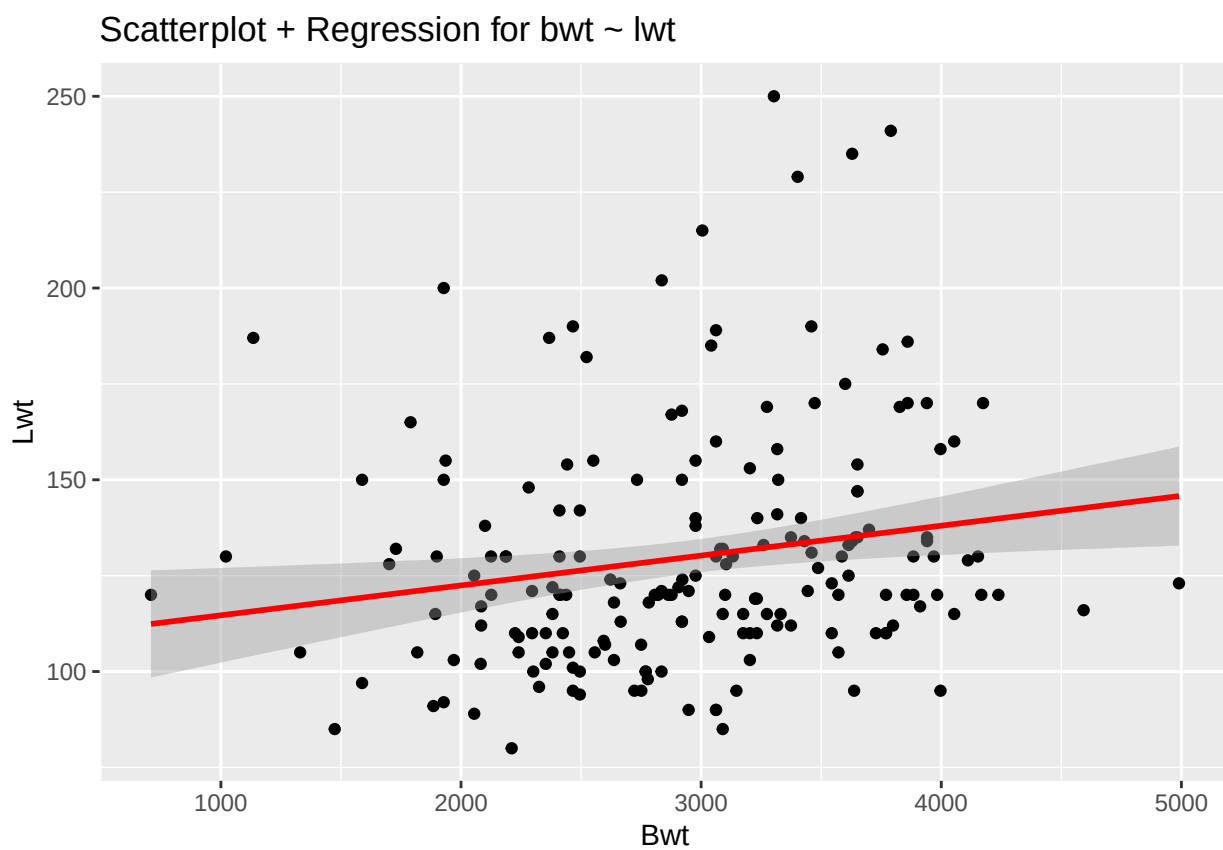
**interpretation of the slope (4.429): On average, the $bwt is estimated to be**

**4.429x as big as the $lwt, e.g. multiplying the $lwt with 4.429 results in an**

**estimation of the respective $bwt**

4. Add the regression line to your scatterplot.

```
ggplot(bwt, aes(x = bwt, y = lwt)) +
  geom_point(colour = "black") +
  geom_smooth(method = "lm", colour = "red") +
  labs(
    title = "Scatterplot + Regression for bwt ~ lwt",
    x = "Bwt",
    y = "Lwt"
  )
```



## Submission checklist (not graded)

- ☐ The file knits without errors.
- ☐ All plots have readable axis labels and titles.
- ☐ Each requested interpretation is written in full sentences.